**-HEMANTH**

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# ELIMINATE DEAD CODE

## Aim:

To write a C program to eliminate dead code from Three Address Code (TAC) by removing statements whose results are not used in the final output.

## Algorithm:

1. start the program.
2. Read Three Address Code statements until END is entered
3. Store the statements and seperate the left hand side (LHS) and right hand side (RHS).
4. Check the statements in reverse order to identify live statements
5. Mark statements whose results are used by other live statements
6. Preserve statements assignes to the output variables x and y.
7. Remove statements that are not required.
8. Display the optimized Three Address Code.
9. Stop the program.

## Pseudocode:

START

READ TAC statements until END

FOR each statement

  Separate LHS and RHS

END FOR

```
FOR each statement in reverse order
    IF statement has no LHS
        Mark it as live
    ELSE IF its result is required
        Mark it as live
    END IF
    IF LHS is x or y
        Mark it as live
    END IF
    IF statement is live
        Mark earlier definitions of its RHS variables
    END IF
END FOR
PRINT all live statements
STOP
```

## Output:

Output

```
Enter Three Address Code statements.
Enter END to finish:
t1 = a + b
t2 = c * d
t3 = t1 + 10
x = t3
y = 20
END

Optimized Three Address Code:
t1 = a + b
t3 = t1 + 10
x = t3
y = 20
```

## Result:

Thus the C program successfully eliminates dead code from Three Address code by removing the unused statement "t2=c*d" and displaying the optimized code